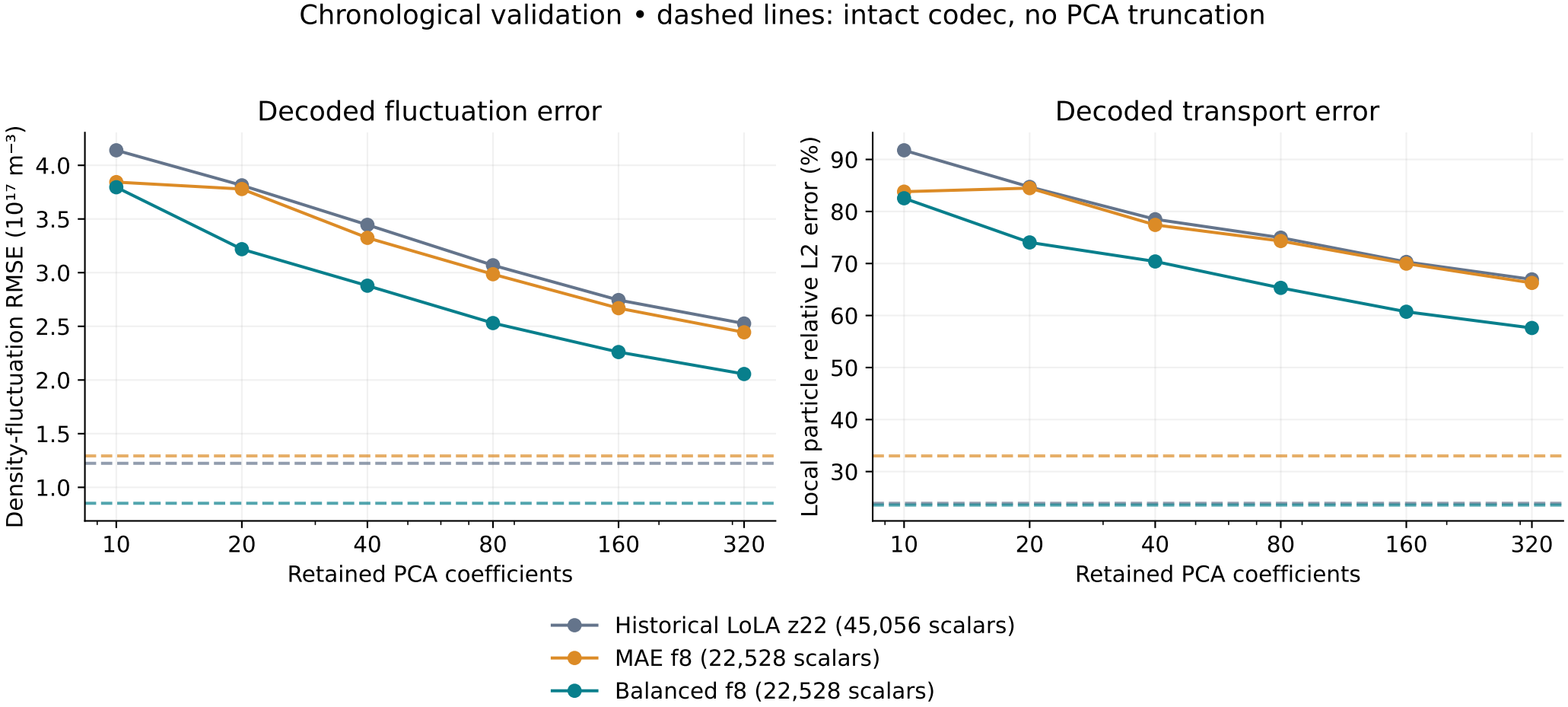
Chronological validation • dashed lines: intact codec, no PCA truncation
Decoded fluctuation error
Density-fluctuation RMSE (10^{17} m^{-3})
4.0
3.5
3.0
2.5
2.0
1.5
1.0
10
20
40
80
160
320
Retained PCA coefficients
Decoded transport error
Local particle relative L2 error (%)
90
80
70
60
50
40
30
10
20
40
80
160
320
Retained PCA coefficients
Historical LoLA z22 (45,056 scalars)
MAE f8 (22,528 scalars)
Balanced f8 (22,528 scalars)